Lateral Transhipment Inventory Models: A Systematic Literature Review of Models and Solution Approaches

Conference Paper · September 2022

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Lateral Transhipment Inventory Models: A Systematic Literature Review of Models and Solution Approaches

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Abstract

Proper inventory management is known to be one of the major challenges in the retail industry. Maintaining a sufficient level of inventory while reducing the total inventory cost results in increasing customer satisfaction and increasing profitability by reducing the overall cost in the retail industry. Lateral transhipment is one of the most effective ways to achieve both objectives. Lateral transhipment is rotating or sharing of stocks among locations in the same echelon. Due to the effectiveness of lateral transhipment in better managing inventory, it has been studied widely in literature. This paper aims to provide a comprehensive review of literature on different lateral transhipment inventory models and solution procedures proposed. The objective is to analyse and identify the research gap for future research and to identify solution approaches used in existing literature for lateral transhipment problems so that the potential research paths can be explored to improve the inventory management in the retail industry. Even though there are reviews conducted for lateral transhipment, to the best of our knowledge most of them do not focus on the solution approaches used. Hence, this paper focuses on the model setting and solution approaches used in the literature. The results of the literature review have revealed that less attention has been given to proactive and hybrid transhipment approaches. Also, the high number of constraints used commonly in lateral transhipment literature can be considered as a restriction in generalising the solutions proposed. It was also concluded that when the number of retailers and items in the considered inventory model increases, approximation techniques were mostly used in literature rather than exact algorithms. Developing complex proactive or hybrid models for multiple item transhipment among multiple retailers or warehouses while considering the allocation of items to vehicles is a future research area to be explored. Also, it was discovered that all the lateral transhipment problems in literature are aligned with economic objectives. Hence, considering the environmental objectives like green approaches is a challenge for future research.

Keywords: Inventory Management, Lateral Transhipment, Retail Industry, Solution Approaches

INTRODUCTION

Product availability is considered as a key success factor in the retail industry. On-shelf availability (OSA) is defined as “the probability of having a product in stock when a customer order arrives” (Chopra & Meindl, 2007). The complement of OSA is a product out of stock (OOS) and it is defined as: “not found in saleable condition, or not shelved in the expected location” (ECR Europe, 2003).

Adverse impacts of OOS are not only limited to the physical store operations but also to the online retail operations. With the advancement of e-commerce retailing, there is high competition among retailers on online platforms as well. The impact of OOS on retailers is higher compared to the impact on manufacturers, due to the increase of options to consumers (Gunness & Oppewal, 2020). Hence to keep up with the competition retails try to increase customer service by ensuring the availability of products at the right time (Grubor & Milicevic, 2015).

Proper inventory management in retail would be beneficial in reducing the stockout occurrences while reducing the total inventory cost. One of the most effective ways to achieve those objectives is Lateral Transshipment. Hence, it has gained a lot of attention from academia and practitioners for decades. Mainly lateral transshipment can be categorized based on the timing of the transshipment: proactive transshipment or reactive transshipment. It can be further classified based on characteristics like the inventory system, ordering behaviour and transshipment type (Paterson et al., 2011).

The complexity of the lateral transshipment problems varies based on the number of locations, number of items, number of echelons and addition of constraints. Hence many authors have used heuristic and meta-heuristic algorithms to find solutions for complex lateral transshipment problems.

The objective of this study is to analyse and identifypossible research gaps for future research and to identify solution approaches that are used in existing literature for lateral transshipment problems. The findings will aid in future to identify the suitable solution (exact mathematical model, heuristic model, or metaheuristic model) approach to develop complex transshipment problems. The comprehensive review of lateral transshipment literature will aid in developing transshipment model to fill the existing research gaps. Even though there have been reviews for lateral transshipment, to the best of our knowledge most of the reviews has not mainly focused on solution approaches used by the authors in solving transshipment problems. Hence, this paper is intended to fill that gap.

METHODOLOGY

The study has used a systematic literature review approach. The systematic review is considered as an unbiased, replicable approach that could be used by practitioners with a reliable basis in decision making (Tranfield et al., 2003).

The methodology of the study is based on the content analysis of lateral transshipment inventory models in the existing literature. The review question of the study was "what are the transshipment models that have been studied in the literature and what are the solution approaches for different models?".

Using keywords, the initial search was conducted to identify the potentially relevant studies. The keywords used are: "lateral transshipment" AND "retail" AND "optimization"; "lateral transshipment" AND "retail" AND "review". Three databases: Google Scholar, Research gate and Science Direct were searched for articles. From this stage, several articles were excluded considering the unavailability of full articles and duplications. The titles and abstract based screening were used as the next step to select the most relevant articles for the full-text screening. Duplicated articles, unrelated articles, abstract-only papers were excluded in this step.

In the full-text screening, the eligibility of the articles was checked. The model setting, language and date were taken as the eligibility criteria. The review has focused on recent journal articles and conference proceeding papers which have been published from the year 2010 in the English language. The rationale for restricting the review to articles published from 2010 is because there is a comprehensive review of lateral transshipment literature done by (Paterson et al., 2011) based on the articles published up to 2010. For the final review, 30 articles were included. The flow diagram of the methodology is shown in figure 1. The article screening and selection process is shown in figure 2.

Figure 33. Flow diagram of the Methodology

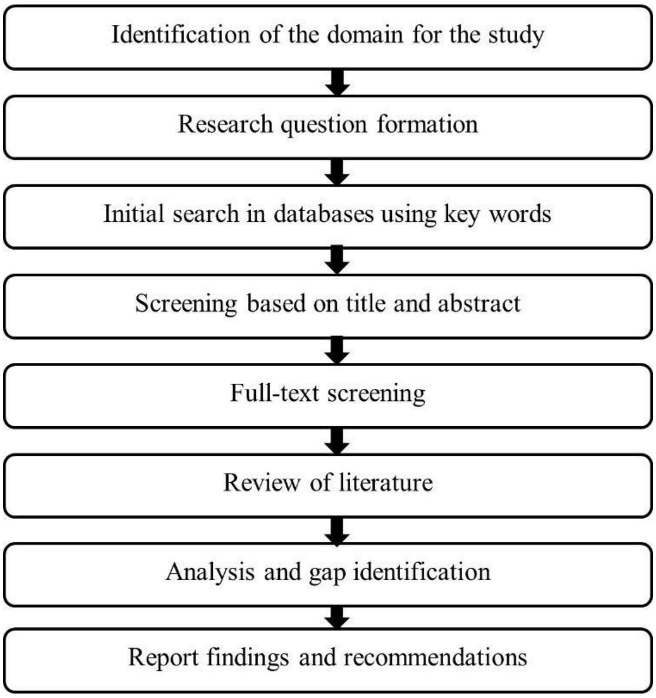
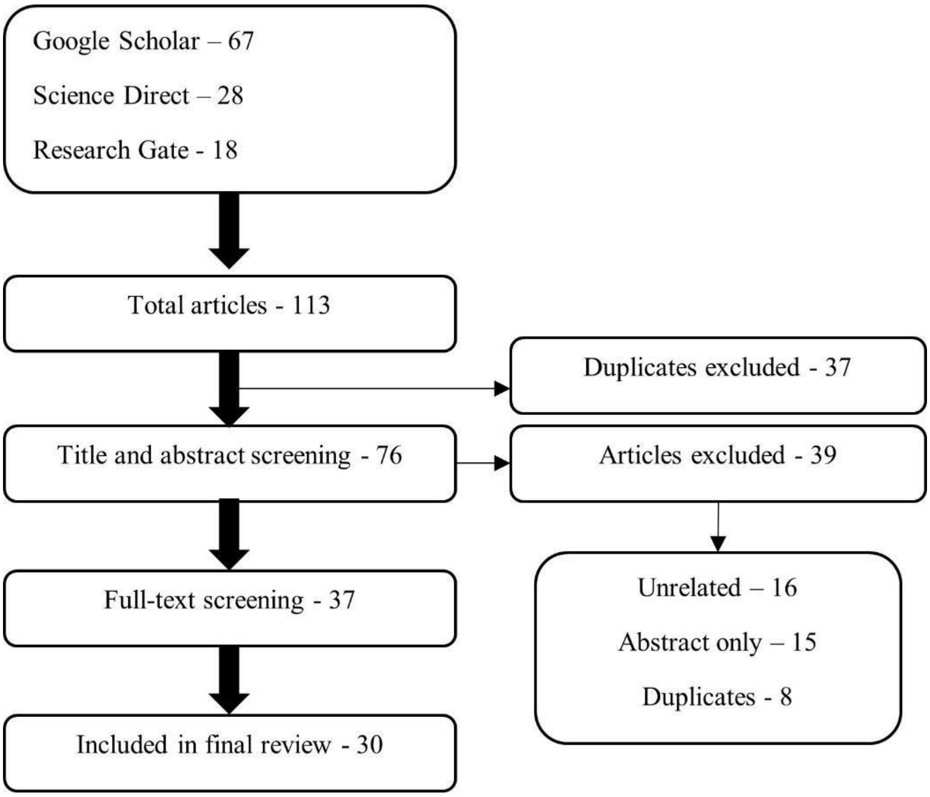


Figure 34. The article screening and selection process



LITERATURE REVIEW

Lateral Transshipment

Lateral transshipment is known as "sharing or rotating stock in a multi-location inventory system at the same echelon in the supply chain" (Nakandala et al, 2017). It has been studied widely in the literature under different constraints for different inventory models. Lateral transshipment has been utilised in many retail industries such as spare parts, sporting goods, apparel, automotive, furniture (Çömez et al., 2012).

Literature in lateral transshipment can be classified based on several criteria. Mainly it is classified depending on the timing of the transshipment as either proactive or reactive. Proactive transshipment or preventive transshipment takes place before the demand realization. At a predetermined point in time, stocks are redistributed amongst all locations in the same echelon (Rim & Jiang, 2019). Proactive transshipment is mostly beneficial in settings where inventory handling cost is high. Because proactive transshipment can be arranged in advance in a way to reduce the handling cost (Paterson et al., 2011).

Reactive transshipment or emergency transshipment takes place once the demand is realized and if one location faces a stockout while another location is having more than sufficient inventory to cover their demand. Reactive transshipment is mostly suitable in instances where the cost of transshipment and inability to respond to customer demand immediately, is less than the inventory holding cost (Paterson et al., 2011). Studies have been conducted which combines both proactive and reactive transshipment types as well. Hence those can be identified as hybrid transshipment models. For example, Nakandala et al. (2017) developed decision rules for a two-echelon inventory system that follows a hybrid transshipment approach.

Paterson et al. (2011) has mentioned many other classification criteria for lateral transshipment. Another classification criterion for reactive transshipment under continuous review is the pooling method: complete or partial. In complete pooling, the transhipping location shares all of its stocks (Paterson et al., 2011). And in partial pooling transshipment happens while the inventory level is above a fixed threshold at the transhipping location (Archibald et al., 2009).

Based on the decision making, lateral transshipment problems can be classified as centralized or decentralized. Centralized transshipment problem is concerned with minimizing the total cost overall stocking points whereas decentralized settings are concerned with maximizing the profitability in each location (Rim & Jiang, 2019).

Lateral transshipment problems can differ from each other depending on the differences in the inventory system as well. They can be classified based on the number of items, number of locations, number of echelons, treatment for unsatisfied demand (backorder or lost sale) and whether the locations are identical (Paterson et al., 2011). In this paper, we have mainly considered the inventory system characteristics to classify the models in the literature.

Reactive Lateral Transshipment

The majority of the transshipment models in the literature have considered reactive lateral transshipment (Abouee-Mehrizi et al., 2015; Meissner & Senicheva, 2018). One of the first studies in reactive transshipment was carried out by Krishnan & Rao (1965) where a single echelon, centralized inventory system was studied under periodic review policy with the objective of cost minimization. After that many authors have studied the reactive transshipment problem under different constraints and inventory models.

Single Period Reactive Transshipment Ekren & Arslan (2020) studied a transshipment problem for a three-location supply network and test four transshipment policies to identify which has the best performance using a simulation approach. The model consists of a single-item one-echelon system and both transporter capacity and lead time have been considered. The results of the simulation have proved that lateral transshipment policy outperforms the option of no transshipment. The authors have used ARENA 14.5 simulation software for the study. Using a simulation approach for the performance analysis of decision rules and models is very common in transshipment literature. It can be identified as a good option since it allows to analyse the validity and performance of models in a more accurate setting.

Paterson et al. (2012) have developed an enhanced reactive transshipment policy that perform better than general reactive policies. In case of a stockout, the policy allows to transfer of additional inventory and hence it has an element of proactive transshipment. The results of a numerical study have shown that gaps are reduced significantly. The authors have used a quasi-myopic approach to solve the transshipment problem.

The spare parts industry can be identified as a main consideration in reactive transshipment (Meissner & Senicheva, 2018). Similar to that, an approximation method for the service part industry to optimize the allocation of stock, has been introduced by Yang et al. (2013) for one or two-echelon systems. . An important aspect of this paper is that it has considered non-negligible transshipment time. The importance of including pipeline stock flexibility along with lateral transshipment in inventory-related decision-making has been proven through the numerical experiment.

The expiry of slow-moving critical medicine is a challenge for the healthcare supply chain. Hence, Jin & Agirbas (2013) has developed a decision rule which takes myopic-best action in each decision period to reduce this issue. And the decision rule has been applied iteratively to each period to determine the cost-saving. The authors have considered a multi-location model where demand arrives in a Poisson distribution. A significant cost saving was shown in the simulation study conducted using the decision rule. The authors have pointed out that if the age of the medicine is tracked properly the decision rule can be applied easily in a real-world scenario. A drawback of the proposed decision rule which uses myopic-best action is the inability to determine the long-run impact based on the current transshipment rule. The authors suggest using the Semi-Markov process to overcome this problem as an extension to this study.

A two-echelon, two retailers centralized system which used periodic review was studied by Hachicha et al. (2013). Using a simulation approach the authors have first determined the optimal values for reorder point and order-up-to value. Later a simulation series has been used to determine the best transshipment policy. The authors have tested multiple transshipment policies and found out that all the transshipment policies provide better results than the no-transshipment policy. One of the concerns of this study is that the findings are only valid under the considered operational assumptions. Also, the simulation study has only considered four different pooling scenarios with no transshipment option. But there can be different ways to do the pooling under different circumstances. Hence the ability to generalize the findings is doubtful.

A markovian model was proposed by Tai & Ching (2014) for an inventory/ return system with two echelons. Lateral transshipment of demand was allowed among multiple warehouses. Numerical analysis and a sensitivity analysis have also been carried out. Some of the operational assumptions and constraints of this study like identical warehouses in terms of cost structure and capacity and first-come-first-served policy has restricted the ability to generalize the model.

Multi-Period Reactive Transshipment: A decentralized inventory system consisting of two competitive retailers was studied by Çömez et al. (2012). These authors have also considered a pairwise optimal holdback levels for two retailers. The authors have delegated the transshipment request acceptance to the source location and the decision will be based on the inventory levels and current time. This research has also considered the customer overflow probability which could be high among competitive retailers in a small geographical area. And the analytical results have shown that transhhipment is beneficial for smaller retailers and geographically distant retailers. The authors have also proposed a heuristic for a multi-retailer setting. It can be noted that this paper has taken a more relaxed approach in solving the transshipment problem by allowing positive transshipment time and multiple transshipment opportunities between two replenishments.

A two-echelon, two retailer system with loss sales and complete pooling which uses (R, s, S) policy was studied by Tlili et al. (2012). An inventory model to determine the best value for (s, S) was developed. A Monte Carlo simulation study has also been conducted by the authors. The study proves that the benefits of transshipment are higher when the lead times are longer. The model can be further improved by increasing the number of retailers. Since the authors have only considered two retailers it does not clearly show the benefit of transshipment when the number of retailers increases.

A setting of two outlets that sell a seasonal product was studied to determine initial inventory and transshipment decisions, by Yao et al. (2016). The authors have emphasized the importance of initial stocking decisions in increasing the profitability of a system. A dynamic programming model has been used to determine the initial stock amounts.

Ramakrishna et al. (2015) have considered a two-item, two warehouse system which allows partial pooling of inventory under periodic review. The model developed by the authors helps to make two decisions: the order-up-to level of each item and the acceptance transshipment request. A greedy heuristics and a Lagrangian relaxation method combined search procedure has been proposed by the authors. The usage of the Lagrangian relaxation method is beneficial in this problem since it allows to expand the solution to a multi-item multi-warehouse setting as well.

A closed-loop supply chain network was studied by Ghomi-avili et al. (2017). The uniqueness of this research is that the authors have considered two resilience strategies for the risk of disruption: extra inventory and lateral transshipment. A mixed-integer programming model has been developed with an objective function to reduce the total cost and it was solved using GAMS 24.3/CPLEX. A sensitivity analysis has been conducted to determine the efficiency of the solution approach.

Proactive Lateral Transshipment

Proactive transhipments allow arranging the transshipment in a way which reduces the handling cost. Paterson et al. (2011) state that proactive transshipment is beneficial in the retail sector where handling cost is dominant. However, proactive transshipment has gotten little attention in the literature compared to reactive transshipment. Allen (1958) is one of the earliest studies in lateral transshipment which considers a proactive transshipment. It has considered the single period, multiple-location setting.

Single Period Proactive Transshipment: A single period proactive transshipment problem with deterministic demand for a fashion retailer was studied by Naderi et al. (2020). The operational constraints of the fashion retailer have made the problem a very complex mixed integer linear problem. As the authors have mentioned a commercial optimizer was not able to solve the problem to optimality effectively. Hence, a heuristic approach was taken to solve the problem. The upper bound for the profit maximization problem was obtained through the Lagrangian Relaxation based approach and lower bounds were obtained using a construction heuristic. To find improved solutions simulated annealing based metaheuristic was used. The authors have shown that constraints like “total number of products a store can send” and the “number of stores that a store can make transshipment” have a significant impact and causes a drastic reduction in gross revenue. The authors have proposed to extend this research to a setting where the demand is uncertain. The high number of operational constraints in this model have made it difficult to be used as a general model while making it a complex model to solve. Especially, constraints like single destination and deterministic demand make its ability to generally use for relaxed inventory models questionable.

Multi-Period Proactive Transshipment: Coelho et al. (2012) has introduced 'inventory routing problem with transshipment (IRPT)' for the situation where vehicle routing and transshipment decision have to be made simultaneously. Due to the complexity and the multiple dimensions in IRPT, it is considered as an NP-hard problem (nondeterministic polynomial hard problem) and a metaheuristic approach was used to obtain the solution. The adaptive large neighbourhood search heuristic has shown that it generates better solutions and the results have revealed that the use of transshipment causes a significant reduction in solution cost.

Extending the research area introduced by Coelho et al. (2012), an inventory routing problem with planned transshipment for a multi-product, multi-period and multi-vehicle system was studied by Peres et al. (2017). The authors have developed an exact formulation as well as a metaheuristic: hybrid Randomized Variable Neighbourhood Descent (RVND). The proposed model has been applied to a large-scale Brazilian retail industry and has shown a significant reduction in logistic cost and inventory levels.

Adding environmental constraints and objectives on top of the economic objectives for the inventory routing problem has been an interesting research area. Hence, an inventory routing problem with a transshipment option was studied by Al-e-hashem & Rekik (2014) with a green approach. The authors have considered a multi-product, multi-period setting and the use of multiple capacitated vehicles in the model. A mixed-integer problem was developed and solved using CPLEX software. The model was developed to choose the appropriate transportation mode and the route in a way that reduces the cost. A sensitivity analysis has also been conducted and authors have mentioned that their model is straightforward to use in practice.

A centralized inventory system consisting of one warehouse and multiple retailers were studied by Feng et al. (2017) under preventive transshipment policy. The authors have divided one review cycle into multiple decision-making periods. The lead time has been considered negligible and the model was developed with an objective function to maximize the profits in remaining decision-making periods. Following the same approach of Archibald et al. (2009) the authors have decomposed the multi-location into pairs of two locations. Then a combined sorting heuristic consisting of backward dynamic programming (BDP) and approximate dynamic programming (ADP) has been used to arrive at the solution. The results have shown the benefit of using proactive transshipment over no transshipment.

The dynamic complexities of a supply chain system with lateral transshipment have been studied by Wei et al. (2018) under a proactive approach. The model consisted of one supplier and two retailers. The impact of lateral transshipment has been analysed using two scenarios:

in the first scenario, the two retailers place orders independently while in the second scenario the order is placed by only one retailer. They have explored the impacts of lateral transshipment on the bullwhip effect, stability, and other performance measurements. The authors have also conducted simulation experiments and the results demonstrate that customer service levels can be improved in a decentralized setting. And based on the results for the two scenarios it was proved that order placement by both retailers is important to keep the supply chain operation effectively.

Meissner & Senicheva (2018) has studied a multi-location and multi-period proactive transshipment model under periodic review. The authors have used an approximate dynamic programming approach to find a near-optimal solution. The developed transshipment policies determine how many items are to be transshipment and the source and destination for the transshipment.

(Dijkstra et al., 2019) has studied cross channel dynamic transshipment policies for a single product in dual channel company. Optimal transshipment policies were gained through the Markov decision process and the authors have introduced a heuristic solution approach as well. The proposed heuristic has performed better than the static policies proving the superiority of dynamic transshipment policies over static policies.

Hybrid Transshipment Approach

The study conducted by Paterson et al. (2012) first introduced the concept of hybrid transshipment where they introduced an enhanced reactive transshipment which transfers additional inventory to reduce future stockouts as well. From there onwards few studies have been conducted which could be considered as hybrid lateral transshipment where both proactive transshipment and reactive transshipment are used in one system.

Single Period Hybrid Transshipment: Nakandala et al. (2017) have studied a two-echelon, multiple location model under periodic review for the hybrid transshipment model. The decision rules developed by the authors help to make trade-off decisions considering different cost elements like holding cost, back ordering cost to choose between backordering and hybrid transshipment option. The authors have considered that the lead time of transshipment is negligible. A numerical study has also been conducted using a simulation approach. The MATLAB software has been used for the calculations, but the authors have mentioned that these decision rules are easy to use and the calculations can be done using spreadsheet software as well. Hence the usability in a real-world business scenario is high. A limitation of the proposed model is that it depends on the previous records of the organization.

A two-echelon inventory system was considered for a hybrid transshipment approach by Ekren et al. (2018). A simulation model has been developed using ARENA 14.0 commercial simulation software. The authors have considered two network designs that differ from each other based on the transportation cost and hub capacity. The results of the simulation have shown that there is a strong correlation between the network design and the performance of the transshipment policy.

Multi-Period Hybrid Transshipment: Following the quasi-myopic approach used by Paterson et al. (2012), a multiple location, multiple item hybrid transshipment model was developed by Glazebrook et al. (2015). The motivation for the model has been taken from a car parts system that follows a periodic replenishment. A simulation study has been conducted to evaluate the performance of the heuristic and the authors have evaluated the hybrid policy, no pooling and complete pooling options. For a large scale network, a Monte Carlo simulation has been carried out. The authors have mentioned that the performance of the hybrid policy increases when the shortage cost is higher. The proposed approach has displayed a significant cost-saving benefit as well as a greener approach in inventory transshipment since fewer vehicle journeys were used for the inventory transshipment.

A multiple item two echelon spare parts distribution system was studied by Topan & van der Heijden (2020). The authors focus on reducing the downtime of the system by integrating reactive and proactive approaches. A mixed integer programming model has been developed and a simulation study has been used to evaluate the performance. For the simulation two types of review: periodic review and opportunistic review have been considered. The authors have also proposed a greedy heuristic approach for the model. It has been observed that in downtime reduction for low priced parts proactive transshipment make a significantly higher contribution. And the combined approach helps to reduce the downtime with a very little increasement in shipment costs. And the authors have mentioned that their proposed model can be used for large scale model settings as well.

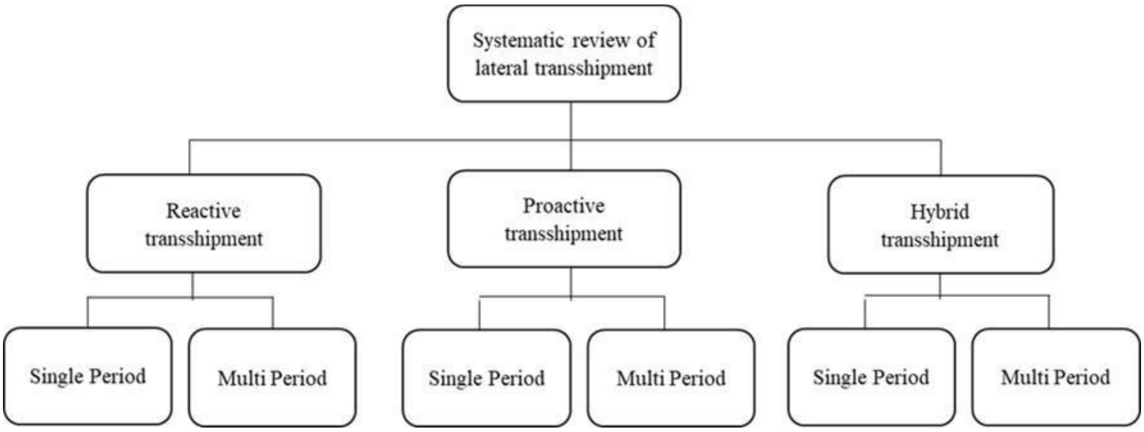
The summary of the literature review is given in table 1. The overview of the literature review is shown in figure 3.

Table 11 Summary of the reviewed articles

	Proactive transshipment	Reactive transshipment	Hybrid transshipment

	Single period	**Multi period**	**Single period**	**Multi period**	**Single period**	**Multi period**
		(Feng et al., 2017)	(Ekren & Arslan, 2020)	(Çömez et al., 2012)	(Nakandala et al., 2017)	
		(Wei et al., 2018)	(Paterson et al., 2012)	(Tlili et al., 2012)	(Ekren et al., 2018)	
		(Coelho et al., 2012)	(Yang et al., 2013)	(Yao et al., 2016)		
		(Meissner & Senicheva, 2018)	(Jin & Agirbas, 2013)	(Nakandala, Lau, & Shum, 2017)		
		(Abouee-Mehrizi et al., 2015)	(Tai & Ching, 2014)	(Emadi & Pasek, 2021)		
		(He et al., 2014)	(Hachicha et al., 2013)			
		(Dijkstra et al., 2019)	(Liao et al., 2020)			
Multi item	(Naderi et al., 2020)	(Al-e-hashem & Rekik, 2014)	(Rim & Jiang, 2019)	(Ghomi-avili et al., 2017)		(Glazebrook et al., 2015)
		(Peres et al., 2017)	(Rim & Vu, 2021)	(Ramakrishna et al., 2015)		(Topan & van der Heijden, 2020)

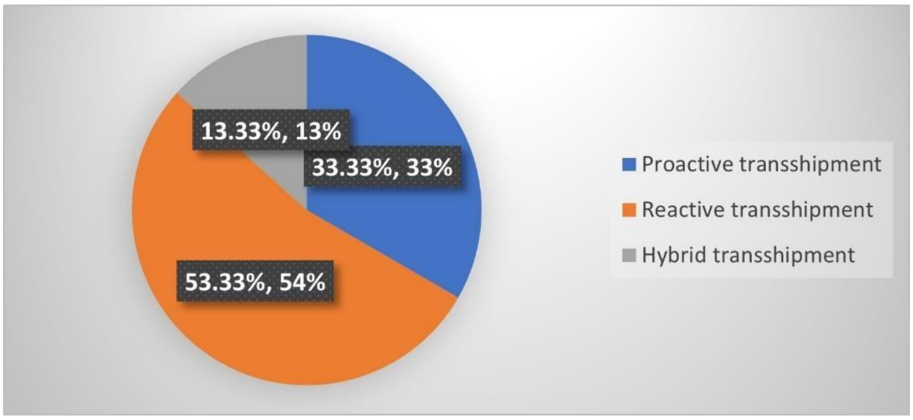
Figure 35. Overview of literature



RESULTS ANALYSIS AND DISCUSSIONS

The literature review was carried out to review the recent literature on lateral transshipment. Mainly the transshipment type, model setting, and the solution approach were analysed. 30 articles were selected for the final review and included in the analysis. The percentages of the transshipment types identified in the review are shown in figure 4.

Figure 36. Percentages of the transshipment types used in the reviewed articles



It can be noted that reactive transshipment has gain more attention in lateral transshipment in the considered period for the review (From the year 2010 to 2021). Comparatively less attention has been given to proactive transshipment. Hybrid transshipment is a combination of proactive transshipment and reactive transshipment. Hybrid transshipment has gained the least attention and this could be considered as a possible research area for future research. And it has been noted that hybrid transshipment provides better results in managing inventory.

Unlike the existing reviews for lateral transshipment, this paper focuses on the solution approaches used by the authors as well. Table 3 provides the summary of the solution approaches used by the reviewed articles. It can be noted that mainly three types of solution approaches were used in the literature. Those are exact mathematical models, heuristic models and metaheuristic models.

It can be noted that exact mathematical models were mainly used by comparatively simple model settings. For example, Liao et al. (2020) has considered a single item single period model and the solution approach is a mathematical model. Whereas some papers have used both the exact mathematical model for a simple and relaxed model setting and heuristic or metaheuristic for more complex settings. For example, Peres et al. (2017). Some of the metaheuristics used are simulated annealing, genetic algorithm, and adaptive large neighbourhood search. The simulation approach has been used by many authors. Some authors have used simulation to evaluate the performance of their proposed models or decision rules. For example, Jin & Agirbas (2013). The model settings and the solution approaches used in each article is shown in table 2.

Table 2. Solution approaches used in the reviewed articles

Research article	Transshipment type	periods	Number of items	Solution approach
(Ekren & Arslan, 2020)	Reactive	Single	Single	Simulation approach
(Paterson et al., 2012)	Reactive	Single	Single	Dynamic programming model
(Yang et al., 2013)	Reactive	Single	Single	An approximation method for optimization of stock allocation
(Jin & Agirbas, 2013)	Reactive	Single	Single	Myopic-best decision rule and simulation
(Tai & Ching, 2014)	Reactive	Single	Single	Markovian queueing model
(Hachicha et al., 2013)	Reactive	Single	Single	Simulation-optimization approach
(Liao et al., 2020)	Reactive	Single	Single	Exact mathematical model
(Rim & Jiang, 2019)	Reactive	Single	Multiple	Linear programming model
(Rim & Vu, 2021)	Reactive	Single	Multiple	Genetic algorithm
(Çömez et al., 2012)	Reactive	Multiple	Single	Exact model and heuristic approach
(Tlili et al., 2012)	Reactive	Multiple	Single	Analytical procedure and

				simulation-optimization model
(Yao et al., 2016)	Reactive	Multiple	Single	Dynamic programming approach
(Nakandala, Lau, & Shum, 2017)	Reactive	Multiple	Single	Heuristic decision rules
(Emadi & Pasek, 2021)	Reactive	Multiple	Single	Deterministic mathematical model
(Ghomi-avili et al., 2017)	Reactive	Multiple	Multiple	Mixed-integer programming
(Ramakrishna et al., 2015)	Reactive	Multiple	Multiple	Combination of greedy heuristics and a Lagrangian relaxation method
(Naderi et al., 2020)	Proactive	Single	Multiple	Mixed-integer programming and Lagrangian relaxation + simulated annealing based metaheuristic
(Coelho et al., 2012)	Proactive	Multiple	Single	Adaptive large neighbourhood search heuristic

Table 2. Solution approaches used in the reviewed articles continued

(Feng et al., 2017)	Proactive	Multiple	Single	Sorting heuristics combined with backward dynamic programming (BDP) approach and approximate dynamic programming (ADP)
(Wei et al., 2018)	Proactive	Multiple	Single	Simulation approach
(Meissner & Senicheva, 2018)	Proactive	Multiple	Single	Approximate dynamic programming
(Al-e-hashem & Rekik, 2014)	Proactive	Multiple	Multiple	Mixed-integer linear programming
(Peres et al., 2017)	Proactive	Multiple	Multiple	Exact formulation and metaheuristic
(Abouee-Mehrizi et al.,	Proactive	Single	Multiple	Dynamic programming

2015)				model
(He et al., 2014)	Proactive	Single	Multiple	Dynamic programming approach
(Dijkstra et al., 2019)	Proactive	Single	Multiple	Markov decision process and heuristic approach
(Nakandala et al, 2017)	Hybrid	Single	Single	Decision rule formation and simulation
(Ekren et al., 2018)	Hybrid	Single	Single	Simulation-based approach
(Glazebrook et al., 2015)	Hybrid	Multipl e	Multiple	Quasi-myopic heuristic
(Topan & van der Heijden, 2020)	Hybrid	Multipl e	Multiple	Mixed-integer programming and Greedy heuristic

Table 3 shows the usage of solution approaches in lateral transshipment literature in different problems. Dynamic programming has been mainly used by studies that have multiple locations systems with the complex setting. And simulation approach has been used by many authors to evaluate the performance of their proposed models or decision rules Mathematical models have been developed for comparatively less complex models.

Note – since some of the articles use more than one algorithm or exact model and heuristic/ metaheuristic approach some articles are considered more than one time in table 3.

Table 3. The usage of solution approaches in lateral transshipment literature

Solution approach	Article	Model setting
Dynamic Programming	(Paterson et al., 2012)	Multi-location inventory system
	(Yao et al., 2016)	A two-location, centralized model
	(Feng et al., 2017)	Multi-location, centralized system with periodic review
	(Meissner & Senicheva, 2018)	Multi-location, centralized system under periodic review
	(Abouee-Mehrizi et al., 2015)	Two-retailer system with stochastic demand, centralized system
	(He et al., 2014)	Two-retailer, centralized system

Simulation-based approach	(Ekren & Arslan, 2020)	One echelon, three stocking location system
	(Jin & Agirbas, 2013)	Multi-location system with slow moving perishable (medical) items
	(Hachicha et al., 2013)	Two-echelon system with a single supplier and two retail locations, periodic review
	(Tlili et al., 2012)	Two-echelon inventory system with one outside supplier, warehouse and two retailers
	(Wei et al., 2018)	Two echelon setting consisted of one external supplier and two retailers, periodic review
	(Ekren et al., 2018)	Two echelons, (s, S) inventory control model
	(Nakandala et al, 2017)	Two-echelon, multi-location system, periodic review
Exact mathematical model	(Liao et al., 2020)	Two stores in two local markets, considers a perishable product
	(Çömez et al., 2012)	Two retailers, Decentralized system
	(Emadi & Pasek, 2021)	Multi-location blood supply chain, perishable item
	(Peres et al., 2017)	Multi- location, multiple vehicle, Inventory Routing Problem with Planned Transshipment (IRPT)
Heuristic models	(Çömez et al., 2012)	Two retailers, Decentralized system
	(Nakandala, Lau, & Shum, 2017)	Perishable inventory, Periodic review
	(Glazebrook et al., 2015)	Multiple locations, Periodic review
	(Dijkstra et al., 2019)	Dual channel company with returned products
Mixed Integer	(Ghomi-avili et al., 2017)	Closed-loop supply chain with

programming		disruption risk
	(Naderi et al., 2020)	Multi-location, centralized system
	(Al-e-hashem & Rekik, 2014)	Inventory Routing Problem (IRP) with multiple vehicles, multiple suppliers and a single plant
	(Topan & van der Heijden, 2020)	Two echelon system with one depot and multiple warehouses, Spare parts, periodic review and opportunistic review

Table 3. The usage of solution approaches in lateral transshipment literature continued

Lagrangian relaxation	(Ramakrishna et al., 2015)	Two-warehouse system with periodic review
	(Naderi et al., 2020)	Multi-location, centralized system
Markovian approach	(Tai & Ching, 2014)	Two echelon system with one supply plant and a warehouse for inventory and returns
	(Dijkstra et al., 2019)	Dual channel company with returned products
Greedy heuristic	(Ramakrishna et al., 2015)	Two-warehouse system with periodic review
	(Topan & van der Heijden, 2020)	Two echelon system with one depot and multiple warehouses, Spare parts, periodic review and opportunistic review
Simulated annealing	(Naderi et al., 2020)	Multi-location, centralized system
Adaptive large neighbourhood	(Coelho et al., 2012)	An inventory routing problem with transshipment
Metaheuristic approach	(Peres et al., 2017)	Multi-location, multiple vehicles, Inventory Routing Problem with Planned Transshipment (IRPT)

Linear programming	(Rim & Jiang, 2019)	Single supplier and multiple DCs, Base stock policy (Order-up-to level)
Genetic algorithm	(Rim & Vu, 2021)	Multiple location, base stock policy, "Transshipment vehicle routing with simultaneous pickup and delivery (TVRSPD)"

Considering the models considered in the reviewed articles, it can be noted that less attention has been given to the transshipment problems of perishable products. Hence it can be identified as a gap for future research. Decentralized systems have also received less attention in lateral transshipment literature.

Table 4 provides the summary of the solution approaches used in the referred articles. It can be noted that the most used solution approaches are the simulation-based approach and dynamic programming.

Table 4. Summary of the solution approaches used in the reviewed articles

Solution approach	Number of articles
Simulation-based approach	07
Dynamic programming	06
Exact mathematical model	04
Heuristic models	04
Mixed-integer programming	04
Lagrangian relaxation	02
Markovian approach	02
Greedy heuristic	02
Simulated annealing	01
Adaptive large neighbourhood	01
Metaheuristic approach	01
Linear programming	01
Genetic algorithm	01
Total	36

It can be noted that many articles have used heuristic or metaheuristic approaches for complex transshipment problems. Some articles have developed an exact model for the relaxed or less complex model setting and after adding some constraints they have used heuristic or metaheuristic approaches. When the dimensionality increases authors have used heuristic or metaheuristic approaches.

Computation time can be considered as an important criterion to evaluate the performance of the algorithms used. But some articles do not provide information regarding the computation time. Hence, it could be taken as a challenge in determining the best performing algorithm. On the other hand, computation time can depend on other criteria like the considered model, the software and the machine used. Hence it is challenging to determine the best performing algorithm for the reviewed literature.

From this review, it was noted that dimensionality can be used as a factor for choosing the solution approach. And if the model setting is complex (for example, multi-location, multi-period, multi-item setting), heuristic or metaheuristic approaches can be used. And decomposition can be used for multi-location problems.

CONCLUSION

The systematic review of literature has focused on recent literature in lateral transshipment which was published in the period; year 2010 to 2021. For the final review, 30 articles were selected through the screening and selection process described under methodology. The objective of this review is to analyse and identify the research gap for future research and to identify solution approaches that are used in existing literature for lateral transshipment problems so that future researchers can identify the suitable solution approach based on the inventory system they consider. Hence, the review was focused on model settings and solution approaches in the recent lateral transshipment literature. The literature was analysed under three main categories. Those are proactive transshipment, reactive transshipment and hybrid transshipment approach. Proactive transshipment has gained comparatively less attention than reactive transshipment due to its complexity. And hybrid transshipment approach can be identified as a research area that could be studied more in future since it has not gained enough attention. Exact mathematical models, heuristics and metaheuristics are used as solution approaches in lateral transshipment literature. It was observed that many of the complex transshipment problems use approximation techniques like heuristic or metaheuristic methods as solution approaches. The exact mathematical models can be used for comparatively less-complex models. Hence, for future research, it is recommended to choose a suitable solution approach considering the complexity and dimensionality of the problem setting. And simulation approach can be used to develop models as well as to evaluate the performance of proposed models. A common questionable aspect of the analysed transshipment models is that they have used multiple constraints to reduce the complexity of the problem which restrict the practical usage and accurate representation of real-world processes. Also, it reduces the ability to generalize the models across industries and settings. Assignment of vehicles for transshipment and lead time consideration for future research is recommended since it has been noted that those were often neglected in the past literature. The proactive or hybrid transshipment of items with short shelf life was identified as a potential future research area. This paper concludes that it is needed to study more about the proactive transshipment or hybrid transshipment of multiple items between multiple stores in the future and for that heuristic or metaheuristic approaches can be chosen to arrive at the solution. Also, the usage of environmental objective functions for lateral transshipment optimization problems have not been explored in literature. Hence lateral transshipment with green approach could be a potential research area for the future. A limitation of this study is the inability of confirming the best algorithm to use due to the lack of information regarding the computation time.

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